MATLAB Drive

Editor

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```
% Comparison of FDMA, TDMA and CDMA
% Simplified illustrative model

clear;
clc;
close all;

% Illustrative system parameters
totalBandwidthHz = 1e6;    % Total bandwidth: 1 MHz
fdmaChannelHz = 200e3;     % FDMA channel width: 200 kHz
tdmaSlots = 8;             % Number of TDMA slots per frame
processingGain = 64;       % CDMA processing gain
targetSIRdB = 7;           % Minimum acceptable CDMA SIR

% Capacity estimation
fdmaUsers = floor(totalBandwidthHz / fdmaChannelHz);

tdmaUsers = tdmaSlots;

targetSIR = 10^(targetSIRdB / 10);

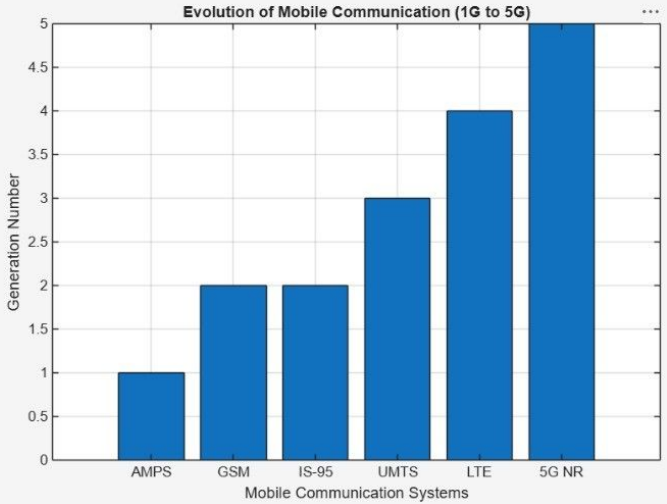
cdmaUsers = floor(1 + processingGain / targetSIR);

% Display maximum supported users
fprintf('Illustrative Maximum Supported Users\n');

fprintf('FDMA: %d users\n', fdmaUsers);
```

Figures

Figure 1



32°C
Mostly sunny

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